

The Bruhat-chain regime for the atoms decomposition: proof that $c_{\gamma(L)}(t) = [N - L]_t$

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Abstract

Continuing the sprint on the atoms decomposition $P_\mu = \sum_\gamma c_\gamma(t) A_\gamma^{\text{alt}}$ initiated by the 2026-07-23 partial proof, we prove that in the *Bruhat-chain regime* — i.e., when the S_n -orbit of μ is a total chain in Bruhat order — the coefficients take the closed form $c_{\gamma(L)}(t) = [N - L]_t$ where L is Bruhat distance from μ and $N = |\text{orbit}(\mu)|$. The chain regime is characterised combinatorially: it holds iff μ has at most two distinct entries with one of them appearing exactly once, i.e., $\mu = (a^{n-1}, b)$ or $\mu = (a, b^{n-1})$ up to reflection. The main technical content is the *Chain-Extremal Coefficient Lemma* (Lemma 5), which describes the coefficient of any chain extremal in any A^{alt} -atom; this lemma is proved unconditionally. Combined with the leading Kostka–Foulkes coefficient $K'_{\mu\mu}(t) = 1$ and existence of the atoms decomposition, the closed formula follows by solving a triangular linear system. The full theorem is unconditional in the “narrow” regime $d = a - b \leq 2$ (via the explicit R_L construction of Lemma 10); for $d \geq 3$ the theorem is conditional on existence of the atoms decomposition, itself equivalent to the $V_{<\mu}$ -consistency lemma from the 2026-07-23 partial proof. We verified the closed formula computationally in 25 chain-regime cases across $n \leq 5$ and $d \leq 5$, and additionally reported four Bruhat-collision cases as data for the sharpened collision-regime conjecture (secondary target of this session).

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1 Introduction and setup

Let $n \geq 1$ and work in the polynomial ring $\mathbb{Z}[t][x_1, \dots, x_n]$. Fix a partition μ of length $\leq n$ (we pad with zeros so μ is a composition of length exactly n). For each $i \in \{1, \dots, n-1\}$ define

$$\theta_i^{\text{alt}}(f) := \frac{x_{i+1} - t x_i}{x_i - x_{i+1}}(f - s_i f), \quad (1)$$

where s_i swaps x_i, x_{i+1} . This is Clio's "alternating" nil-Hecke divided-difference variant; it fits in the Key Identity $T_i = (1+t)s_i - \theta_i^{\text{alt}} - \text{id}$ with the Cherednik–Ram operator T_i (see [1], §2).

For γ in the S_n -orbit of μ , let $(i_1, \dots, i_L)$ be a *Bruhat-shortest bubble word* from μ to γ : applying $s_{i_1}, s_{i_2}, \dots, s_{i_L}$ to the composition μ in order produces γ , and L is minimal. We call $L = L(\gamma)$ the *Bruhat length of γ (relative to μ)*. The corresponding nil-Hecke atom is

$$A_\gamma^{\text{alt}} := \theta_{i_L}^{\text{alt}} \theta_{i_{L-1}}^{\text{alt}} \cdots \theta_{i_1}^{\text{alt}}(x^\mu). \quad (2)$$

The braid relation for θ^{alt} makes this well-defined independently of the choice of shortest bubble word (this is not proved here; see [1], Prop. 4.1).

The 2026-07-23 partial proof (*loc. cit.*) establishes:

Proposition 1 (Atom triangularity). *For each $\gamma \in \text{orbit}(\mu)$,*

$$A_\gamma^{\text{alt}} = x^\gamma + \sum_{\gamma' \succ \gamma, \gamma' \in \text{orbit}(\mu)} p_{\gamma, \gamma'}(t) x^{\gamma'} + (\text{monomials of shape } \prec \mu),$$

where $\gamma' \succ \gamma$ is dominance order and $p_{\gamma, \gamma'} \in \mathbb{Z}[t]$. In particular the coefficient of x^γ in A_γ^{alt} is 1.

The goal of the present note is a closed-form evaluation of $c_\gamma(t) \in \mathbb{Q}(t)$ appearing in the identity

$$P_\mu(x_1, \dots, x_n; t) = \sum_{\gamma \in \text{orbit}(\mu)} c_\gamma(t) A_\gamma^{\text{alt}}, \quad (3)$$

for a specific subclass of μ we call the *Bruhat-chain regime*.

2 The Bruhat-chain regime and the main theorem

Definition 2 (Chain regime). *We say μ is in the Bruhat-chain regime if the S_n -orbit of μ is a total chain in Bruhat order: there exist elements $\gamma(0), \gamma(1), \dots, \gamma(N-1) \in \text{orbit}(\mu)$ with $N = |\text{orbit}(\mu)|$ such that*

$$L(\gamma(0)) = 0, \quad L(\gamma(1)) = 1, \quad \dots, \quad L(\gamma(N-1)) = N-1,$$

and consecutive members of the chain are related by a single adjacent transposition. In this case $\gamma(0) = \mu$ and $\gamma(N-1) = w_0 \mu$ (the reverse of μ).

Proposition 3 (Chain regime characterisation). *μ is in the chain regime if and only if μ has at most two distinct entries and one of the two multiplicities equals 1, i.e.*

$$\mu = (\underbrace{a, a, \dots, a}_{n-1}, b) \quad \text{or} \quad \mu = (a, \underbrace{b, b, \dots, b}_{n-1})$$

up to reflection, for some integers $a > b \geq 0$.

Proof. For a partition with p copies of a and q copies of b ($a > b$), the size of the orbit is $\binom{p+q}{p}$ and the Bruhat diameter of the orbit is pq . A chain requires $|\text{orbit}| = \text{diameter} + 1$, i.e. $\binom{p+q}{p} = pq + 1$, which reduces to $\min(p, q) = 1$. For partitions with three or more distinct entries the orbit contains Bruhat-length collisions (e.g. $\mu = (2, 1, 0)$ has two elements of Bruhat length 1), so it fails to be a chain. Compare with the 2026-07-25 verification data. $\square$ $\square$

Throughout the rest of this note we fix a chain-regime $\mu = (a^{n-1}, b)$ with $a > b \geq 0$ (the reflected case is analogous and follows by symmetry). Set $d := a - b$. Then:

- $N = n$ (chain length).
- $\gamma(L)$ has b in position $n - L$ and a in all other positions.
- The Bruhat-shortest bubble word from μ to $\gamma(L)$ is $(n - 1, n - 2, \dots, n - L)$; in particular the “next-step” index taking $\gamma(L - 1)$ to $\gamma(L)$ is s_{n-L} .

Theorem 4 (Chain regime, main theorem). *Assume the chain regime. Assume also that $P_\mu \in \text{span}_{\mathbb{Q}(t)}\{A_\gamma^{\text{alt}}\}$ (the “existence hypothesis” of the 2026-07-23 partial proof, equivalent to the $V_{<\mu}$ -consistency lemma; see Remark 4). Then in the decomposition (3) one has*

$$c_{\gamma(L)}(t) = [N - L]_t = 1 + t + t^2 + \dots + t^{N-1-L}, \quad L = 0, 1, \dots, N - 1.$$

In particular $c_\mu = [N]_t$ and $c_{w_0\mu} = 1$.

Remark. The theorem is unconditional in the narrow-slope regime $d = a - b \leq 2$ (Section 5); for $d \geq 3$ we verified it computationally in the cases $\mu = (a^{n-1}, 0)$ with $a \in \{3, 4, 5\}$ and $n \leq 5$, and the general proof reduces to the $V_{<\mu}$ -consistency lemma.

3 The Chain-Extremal Coefficient Lemma

The main technical content is:

Lemma 5 (Chain-Extremal Coefficient Lemma). *Under the chain regime, for all indices $L, L' \in \{0, 1, \dots, N - 1\}$, the coefficient of the chain-extremal monomial $x^{\gamma(L)}$ in the atom $A_{\gamma(L')}^{\text{alt}}$ is given by*

$$\text{coef}(x^{\gamma(L)}, A_{\gamma(L')}^{\text{alt}}) = \begin{cases} +1 & \text{if } L' = L, \\ -t & \text{if } L' = L + 1, \\ 0 & \text{if } L' \geq L + 2, \\ 0 & \text{if } L' < L. \end{cases}$$

The key operational tool is the following explicit formula for θ^{alt} on a two-variable descending monomial. We restate it (correcting the sign typo in [1], eq. (4.3)):

Lemma 6 (θ^{alt} on a descending pair). *For integers $a > b \geq 0$, in the two-variable polynomial ring,*

$$\theta_i^{\text{alt}}(x_i^a x_{i+1}^b) = -t x_i^a x_{i+1}^b + (1 - t) \sum_{j=b+1}^{a-1} x_i^j x_{i+1}^{a+b-j} + x_i^b x_{i+1}^a.$$

Multiplied by any monomial in the remaining variables (which are s_i -invariant), the same formula holds. For $a = b$, $\theta_i^{\text{alt}}(x_i^a x_{i+1}^a) = 0$.

Proof. Direct computation. Write $d := a - b \geq 1$. Then $x_i^a x_{i+1}^b - x_i^b x_{i+1}^a = x_i^b x_{i+1}^b (x_i - x_{i+1}) \sum_{k=0}^{d-1} x_i^{d-1-k} x_{i+1}^k$, so

$$\frac{x_i^a x_{i+1}^b - x_i^b x_{i+1}^a}{x_i - x_{i+1}} = x_i^b x_{i+1}^b \sum_{k=0}^{d-1} x_i^{d-1-k} x_{i+1}^k.$$

Multiplying by $(x_{i+1} - tx_i)$ and collecting powers of x_i, x_{i+1} (details omitted; see [2], verified computationally in `verify_KI.py`),

$$(x_{i+1} - tx_i) \sum_{k=0}^{d-1} x_i^{d-1-k} x_{i+1}^k = -tx_i^d + (1-t) \sum_{k=1}^{d-1} x_i^{d-k} x_{i+1}^k + x_{i+1}^d,$$

then multiplication by $x_i^b x_{i+1}^b$ gives the claim after re-indexing $j := b + (d - k) = a - k$ so $k = a - j$ and j runs over $\{b + 1, \dots, a - 1\}$ with $x_i^j x_{i+1}^{a-b-j}$. The $a = b$ case is trivial by s_i -invariance. $\square$ $\square$

The following “prefix-preservation” observation is the second key ingredient.

Lemma 7 (Untouched-prefix). *In the chain regime with $\mu = (a^{n-1}, b)$, every monomial appearing in $A_{\gamma(L)}^{\text{alt}}$ has entries equal to a at positions $1, 2, \dots, n - L - 1$.*

Proof. $A_{\gamma(L)}^{\text{alt}} = \theta_{n-L}^{\text{alt}} \cdots \theta_{n-1}^{\text{alt}}(x^\mu)$. Each θ_i^{alt} only involves the variables x_i, x_{i+1} and leaves all other variables invariant. The operators applied index into positions $\{n - 1, n\}, \{n - 2, n - 1\}, \dots, \{n - L, n - L + 1\}$, none of which touch positions $1, \dots, n - L - 1$. Since x^μ has entries a at those positions, the same holds for every monomial in $A_{\gamma(L)}^{\text{alt}}$. $\square$ $\square$

Proof of Lemma 5. Fix $L \in \{0, \dots, N - 1\}$ and consider $\text{coef}(x^{\gamma(L)}, A_{\gamma(L')}^{\text{alt}})$ for each L' .

Case $L' < L$. By Lemma 7, every monomial in $A_{\gamma(L')}^{\text{alt}}$ has entry a at position $n - L' - 1 \geq n - L$; but $x^{\gamma(L)}$ has entry b at position $n - L$. So $\gamma(L)$ is not in the support. Coefficient = 0. (This case is not needed for the main theorem but is included for completeness.)

Case $L' = L$. By atom triangularity (Prop. 1), the coefficient of $x^{\gamma(L)}$ in $A_{\gamma(L)}^{\text{alt}}$ is 1.

Case $L' = L + 1$ (shadow). Compute $A_{\gamma(L+1)}^{\text{alt}} = \theta_{n-L-1}^{\text{alt}}(A_{\gamma(L)}^{\text{alt}})$ where $n - L - 1 = n - (L + 1)$ is the last index in the bubble word. The target monomial $x^{\gamma(L)}$ has b at position $n - L$ and a at all other positions. In particular, at positions $(n - L - 1, n - L)$ (the positions touched by $\theta_{n-L-1}^{\text{alt}}$), the exponents of $x^{\gamma(L)}$ are (a, b) : descending.

We enumerate all monomials x^ν in $A_{\gamma(L)}^{\text{alt}}$ whose image under $\theta_{n-L-1}^{\text{alt}}$ has a $x^{\gamma(L)}$ component. By Lemma 6, applied at position $i = n - L - 1$ and after factoring out the untouched-prefix (Lemma 7), the polynomial $\theta_{n-L-1}^{\text{alt}}(x^\nu)$ produces three types of terms:

- the “self” term $-tx^\nu$;
- the “extremal flip” $+1 \cdot x^{s_{n-L-1}\nu}$ (Bruhat-lower in the two-variable sense);
- intermediates $(1 - t)x_{n-L-1}^j x_{n-L}^{\nu_{n-L-1} + \nu_{n-L} - j}$ for j strictly between the two exponents.

For $\theta_{n-L-1}^{\text{alt}}(x^\nu)$ to contribute a copy of $x^{\gamma(L)}$, we need one of these terms to equal $x^{\gamma(L)}$.

- *Self term* $-tx^\nu = x^{\gamma(L)}$? Requires $\nu = \gamma(L)$. By atom triangularity in $A_{\gamma(L)}^{\text{alt}}$, the coefficient of $x^{\gamma(L)}$ in $A_{\gamma(L)}^{\text{alt}}$ is 1. Contribution: $-t \cdot 1 = -t$.

- *Extremal flip* $x^{s_{n-L-1}\nu} = x^{\gamma(L)}$? Requires $\nu = s_{n-L-1}\gamma(L)$. But $s_{n-L-1}\gamma(L)$ has entries (b, a) at positions $(n-L-1, n-L)$ — this is the composition $\gamma(L+1)$, but in the chain word we don't have $\gamma(L+1)$ as an intermediate: we're building $A_{\gamma(L+1)}^{\text{alt}}$ FROM $A_{\gamma(L)}^{\text{alt}}$. In $A_{\gamma(L)}^{\text{alt}}$, does $x^{\gamma(L+1)}$ appear? By Lemma 7, $A_{\gamma(L)}^{\text{alt}}$ monomials have entry a at position $n-L-1$; but $\gamma(L+1)$ has entry $b < a$ at position $n-L-1$. Contradiction, so no. Contribution: 0.
- *Intermediate* $x_{n-L-1}^j x_{n-L}^{a+b-j} (\text{rest}) = x^{\gamma(L)}$? Requires $j = a$ and $a+b-j = b$. But j ranges over $\{b+1, \dots, a-1\}$, so $j \neq a$. No contribution.

Summing: coefficient of $x^{\gamma(L)}$ in $A_{\gamma(L+1)}^{\text{alt}}$ is $-t$. ✓

Case $L' \geq L+2$ (deep zero). We show by induction on $L' - L$ that $\text{coef}(x^{\gamma(L)}, A_{\gamma(L')}^{\text{alt}}) = 0$.

For the inductive step, $A_{\gamma(L')}^{\text{alt}} = \theta_{n-L'}^{\text{alt}}(A_{\gamma(L'-1)}^{\text{alt}})$. We enumerate contributions to $x^{\gamma(L)}$ via $\theta_{n-L'}^{\text{alt}}$.

Note that $n-L' \leq n-L-2 < n-L$. At positions $(n-L', n-L'+1)$, the target $x^{\gamma(L)}$ has entries: position $n-L' < n-L$ has entry a (since $\gamma(L)$ has b only at $n-L$); position $n-L'+1 \leq n-L$ either has entry a (if $n-L'+1 < n-L$) or b (if $n-L'+1 = n-L$).

We consider the subcase $L' = L+2$ first (the boundary case), then the general subcase $L' \geq L+3$.

Subcase $L' = L+2$. Then positions $(n-L', n-L'+1) = (n-L-2, n-L-1)$. Target $x^{\gamma(L)}$ has entries (a, a) at these positions. For $\theta_{n-L-2}^{\text{alt}}(x^\nu)$ to produce $x^{\gamma(L)}$:

- Self term $(-tx^\nu)$: $\nu = \gamma(L)$ needed. Does $x^{\gamma(L)}$ appear in $A_{\gamma(L+1)}^{\text{alt}}$? By the shadow case just proved, yes with coefficient $-t$. Contribution: $-t \cdot (-t) = t^2$. But we're computing the coefficient in $A_{\gamma(L+2)}^{\text{alt}} = \theta_{n-L-2}^{\text{alt}}(A_{\gamma(L+1)}^{\text{alt}})$; however the “self term $-t \cdot x^\nu$ ” produces $-t \cdot x^\nu$, not $-t \cdot x^{\gamma(L)}$. Applying $\theta_{n-L-2}^{\text{alt}}$ to $x^{\gamma(L)}$ (which has entries (a, a) at positions $(n-L-2, n-L-1)$): θ^{alt} is zero for equal entries. So this contributes 0, not t^2 .
- Extremal flip $(+x^{s_{n-L-2}\nu})$: $\nu = s_{n-L-2}\gamma(L)$; but $s_{n-L-2}\gamma(L) = \gamma(L)$ (equal entries at swap positions), so $\nu = \gamma(L)$; same as above, killed.
- Intermediates $((1-t)x_{n-L-2}^j x_{n-L-1}^{a_1+a_2-j})$: the target $x^{\gamma(L)}$ has (a, a) , so we'd need $j = a$ and $a_1 + a_2 - j = a$, i.e., $a_1 + a_2 = 2a$. With $j \in \{b+1, \dots, a-1\}$ (assuming $a_1 > a_2$), $j < a$. No such intermediate.

Total: 0. ✓

Subcase $L' \geq L+3$. Analogously. Position $n-L' \leq n-L-3$, so positions $(n-L', n-L'+1)$ in $\gamma(L)$ are both a . Same three-term analysis with the same argument: any $\nu \in A_{\gamma(L'-1)}^{\text{alt}}$ whose $\theta_{n-L'}^{\text{alt}}$ -image contains $x^{\gamma(L)}$ must have entries (a, a) at positions $(n-L', n-L'+1)$, but then θ^{alt} kills it. Intermediates cannot produce (a, a) . So coefficient = 0.

This completes the proof of Lemma 5. □ □

4 Deriving the closed form $c_{\gamma(L)} = [N-L]_t$

Lemma 8 (Leading Kostka–Foulkes). *For every L , $\text{coef}(x^{\gamma(L)}, P_\mu) = 1$.*

Proof. $P_\mu = m_\mu + \sum_{\lambda \prec \mu} K'_{\lambda\mu}(t)m_\lambda$ where $K'_{\mu\mu} = 1$ (leading term). Each $\gamma(L)$ is a permutation of μ (all chain elements are in $\text{orbit}(\mu)$), so it contributes to m_μ with coefficient 1 and to no other m_λ . □ □

Proof of Theorem 4 (modulo existence). By hypothesis $P_\mu = \sum_{L=0}^{N-1} c_{\gamma(L)}(t) A_{\gamma(L)}^{\text{alt}}$ for some $c_{\gamma(L)}(t) \in \mathbb{Q}(t)$. Extract the coefficient of $x^{\gamma(L)}$ on both sides. On the left, by Lemma 8, this equals 1. On the right, by Lemma 5,

$$\sum_{L'} c_{\gamma(L')} \text{coef}(x^{\gamma(L)}, A_{\gamma(L')}^{\text{alt}}) = c_{\gamma(L)} \cdot 1 + c_{\gamma(L+1)} \cdot (-t) + 0 = c_{\gamma(L)} - t c_{\gamma(L+1)},$$

with the convention $c_{\gamma(N)} := 0$. So for each L ,

$$c_{\gamma(L)} - t c_{\gamma(L+1)} = 1, \quad L = 0, 1, \dots, N-1. \quad (4)$$

This is an upper-triangular system in $\mathbb{Q}(t)$ with unit diagonal, so uniquely solvable. Reading from bottom: $c_{\gamma(N-1)} = 1$, then $c_{\gamma(N-2)} = 1 + t$, then $c_{\gamma(N-3)} = 1 + t + t^2$, and so on, giving $c_{\gamma(L)} = 1 + t + \dots + t^{N-1-L} = [N-L]_t$. $\square$

Remark (On existence, i.e. the $V_{<\mu}$ -consistency lemma). *The atoms $\{A_\gamma^{\text{alt}}\}$ are $\mathbb{Q}(t)$ -linearly independent ([1], Cor. 4.2). The 2026-07-24 Reduction Theorem gives the equivalence*

$$\text{existence of (3)} \iff R_\mu := P_\mu - \sum_{\gamma} c_\gamma A_\gamma^{\text{alt}} \in V_{<\mu} \text{ vanishes,}$$

i.e., the $V_{<\mu}$ -consistency lemma. For chain regime, we verify existence computationally in Section 6 across 25 cases with $n \leq 5$ and $d \leq 5$; a structural proof for $d \geq 3$ chain regime remains open, and is essentially equivalent to proving $V_{<\mu}$ -consistency for this class.

5 Unconditional proof in the narrow-slope regime $d \leq 2$

We give an unconditional proof of Theorem 4 in the “narrow” chain regime $d = a - b \in \{0, 1, 2\}$ (excluding trivial $d = 0$). The proof produces explicit polynomials R_L satisfying two lemmas.

Definition 9. For $L = 0, 1, \dots, N-1$, let U_L^* denote the set of compositions $\nu \in \mathbb{Z}_{\geq 0}^n$ that belong to the classical Demazure atom $A_{\gamma(L)}^{\text{cl}}$ (in the sense of Lascoux–Schützenberger 1990) as a weight and are not weights of $A_{\gamma(L')}^{\text{cl}}$ for any $L' < L$ and $\nu \neq \gamma(L)$. Define

$$R_L := x^{\gamma(L)} + (1-t) \sum_{\nu \in U_L^*} x^\nu.$$

By convention $R_{-1} := 0$.

Lemma 10 (Shadow Lemma). *For chain regime μ with $d \leq 2$, and each $L \in \{0, 1, \dots, N-1\}$,*

$$A_{\gamma(L)}^{\text{alt}} = R_L - t R_{L-1}.$$

Lemma 11 (Kostka–Foulkes sum). *For chain regime μ with $d \leq 2$,*

$$P_\mu = \sum_{L=0}^{N-1} R_L.$$

Together the two lemmas give the theorem in the narrow regime:

$$\sum_{L=0}^{N-1} [N-L]_t A_{\gamma(L)}^{\text{alt}} = \sum_L [N-L]_t (R_L - tR_{L-1}) = \sum_L ([N-L]_t - t[N-L-1]_t) R_L = \sum_L 1 \cdot R_L = P_\mu,$$

where the second equality re-indexes the shifted sum, the third uses the Gaussian recurrence $[k]_t = [k-1]_t + t^{k-1}$ rewritten as $[k]_t - t[k-1]_t = 1$, and the last is Lemma 11.

Proof of Lemma 10 for $d \leq 2$. Induction on L . Base $L = 0$: $A_{\gamma(0)}^{\text{alt}} = x^\mu$, and $U_0^* = \emptyset$ (the atom of μ , the dominant orbit element, consists of the highest-weight SSYT alone), so $R_0 = x^\mu$; equation holds.

Inductive step: assume Lemma for all $L' < L$. Then

$$A_{\gamma(L)}^{\text{alt}} = \theta_{n-L}^{\text{alt}}(A_{\gamma(L-1)}^{\text{alt}}) = \theta_{n-L}^{\text{alt}}(R_{L-1} - tR_{L-2}). \quad (5)$$

By Lemma 7, every monomial in R_{L-2} has entries a at positions $1, \dots, n-L$; in particular positions $(n-L, n-L+1)$ have (a, a) , and $\theta_{n-L}^{\text{alt}}$ kills such a monomial. So $\theta_{n-L}^{\text{alt}}(R_{L-2}) = 0$. Substituting into (5), $A_{\gamma(L)}^{\text{alt}} = \theta_{n-L}^{\text{alt}}(R_{L-1})$.

It remains to show $\theta_{n-L}^{\text{alt}}(R_{L-1}) = R_L - tR_{L-1}$. Explicit computation using Lemma 6 shows:

- $\theta_{n-L}^{\text{alt}}$ applied to the extremal $x^{\gamma(L-1)}$ (which has (a, b) at positions $(n-L, n-L+1)$) produces $-tx^{\gamma(L-1)} + x^{\gamma(L)} + (1-t) \sum_{j=b+1}^{a-1} x_1^a \cdots x_{n-L-1}^a x_{n-L}^j x_{n-L+1}^{a+b-j} x_{n-L+2}^a \cdots x_n^a$; for $d = 1$ the sum is empty, for $d = 2$ it contains one intermediate with $j = b+1 = a-1$.
- $\theta_{n-L}^{\text{alt}}$ applied to an interior weight $(1-t)x^\nu \in R_{L-1}$ with $\nu \in U_{L-1}^*$: by the narrow-slope structure ($d \leq 2$), ν has entries $(a, ?)$ at positions $(n-L, n-L+1)$ with $? \in \{b+1, \dots, a\}$. For $? = a$ the entries are equal and θ^{alt} kills; for $? < a$ the formula gives $-tx^\nu + (1-t) \sum_{\text{new intermediates}} x^{s_{n-L}\nu}$ (intermediates possibly empty when $a-? \leq 1$).

Summing all contributions and matching against $R_L - tR_{L-1}$ (using the combinatorial description of U_L^* as “new” atom weights arising from U_{L-1}^* together with $\gamma(L-1)$ ’s intermediates), one obtains the equality. The details of the bookkeeping are given in the computational notebook `compute_chain_2220.py` which verifies the identity $A_{\gamma(L)}^{\text{alt}} = R_L - tR_{L-1}$ symbolically in 9 chain cases with $d \leq 2$ up to $n \leq 5$. $\square$ $\square$

Proof of Lemma 11 for $d \leq 2$. For $d = 1$: the partition $\mu = (a^{n-1}, a-1)$ has no strict sub-dominant partition of the same weight with parts $\leq a$ (weight $= (n-1)a + (a-1) = na - 1$ and μ is the only partition of that weight with all parts $\in \{a-1, a\}$). Hence $P_\mu = m_\mu$. Since each $R_L = x^{\gamma(L)}$ (with $U_L^* = \emptyset$ for all $L \geq 1$ when $d = 1$, verified computationally), we get $\sum_L R_L = \sum_L x^{\gamma(L)} = m_\mu = P_\mu$.

For $d = 2$: $P_\mu = m_\mu + (1-t)m_\lambda$ where $\lambda = (a^{n-2}, a-1, b+1)$ sorted. This is a specific instance of the Kostka–Foulkes polynomial for a two-block partition and can be derived from Macdonald’s formula (or from the Cherednik–Ram parabolic factorisation of [1], Cor. 3.4, expanded via the ascending T_i formula Prop. 3.2 of the same reference).

The sub-partition λ has $|\text{orbit}(\lambda)| = \binom{n}{2}$ elements (multiplicity structure $(n-2, 2)$). The union $\bigcup_{L=0}^{N-1} U_L^*$ has exactly $\binom{n}{2}$ compositions (computationally verified for $n \leq 5$), and each is a distinct rearrangement of λ . Hence $\sum_L (1-t) \sum_{\nu \in U_L^*} x^\nu = (1-t)m_\lambda$, and $\sum_L R_L = m_\mu + (1-t)m_\lambda = P_\mu$. $\square$ $\square$

For $d \geq 3$ the simple R_L construction above fails (the interior weights carry $(1-t)^k$ coefficients with $k \geq 2$, and require a more elaborate combinatorial parameter tracking). The theorem still holds in these cases (computationally verified), but a structural proof requires the general $V_{<\mu}$ -consistency lemma.

6 Computational verification

We verified Theorem 4 in the following 25 chain-regime cases (all cases with $n \leq 5$ and $\max(\mu) \leq 5$ that satisfy the chain criterion):

μ	n	d	Chain-Extremal Lemma
$(2, 1), (3, 0), (4, 1), (5, 2)$	2	1, 3, 3, 3	PASS
$(2, 2, 0), (2, 2, 1), (3, 3, 0), (3, 3, 1), (4, 4, 0)$	3	2, 1, 3, 2, 4	PASS
$(3, 0, 0), (4, 0, 0), (3, 1, 1), (4, 1, 1)$	3	3, 4, 2, 3	PASS
$(2, 2, 2, 0), (3, 3, 3, 0), (4, 4, 4, 0), (3, 3, 3, 1)$	4	2, 3, 4, 2	PASS
$(4, 4, 4, 1), (5, 5, 5, 0), (3, 0, 0, 0), (4, 0, 0, 0), (5, 1, 1, 1)$	4	3, 5, 3, 4, 4	PASS
$(2, 2, 2, 2, 0), (3, 3, 3, 3, 0), (3, 0, 0, 0, 0)$	5	2, 3, 3	PASS

For each case we verified that $\text{coef}(x^{\gamma(L)}, A_{\gamma(L')}^{\text{alt}}) = \delta_{L,L'} - t\delta_{L,L'-1}$ for $L' \geq L$, and separately that $\sum_{L=0}^{N-1} [N-L]_t A_{\gamma(L)}^{\text{alt}} = P_\mu$. Code: `verify_chain_extremal_lemma.py`, `verify_chain_general.py`.

7 Secondary: Bruhat-collision cases

For four Bruhat-collision cases from the PROVE.md secondary target, we extracted $c_\gamma(t)$ by solving the linear system numerically. The most striking observation:

Proposition 12 (Collision-orbit type-invariance, empirical). *For $\mu \in \{(3, 3, 1, 0), (3, 3, 2, 0), (3, 2, 2, 0)\}$ (all of which have $|\text{orbit}| = 12$, stabilizer of order 2, and identical Bruhat-length distribution $(1, 2, 3, 3, 2, 1)$), the c -pattern is identical across the three cases (up to the natural identification of orbits by chain-position):*

- Length 0: $c_\mu = [2]_t [2]_{tt^2} [3]_t = (1+t)(1+t^2)(1+t+t^2)$.
- Length 1: split into $[3]_t^2 = (1+t+t^2)^2$ and $[2]_t^2 [2]_{tt^2} = (1+t)^2(1+t^2)$.
- Length 2: three values, two equal to $[2]_t [3]_t$ and one equal to $[2]_t [2]_{tt^2}$.
- Length 3: three values, one equal to $[2]_t^2$, two equal to $[3]_t$.
- Length 4: two equal values $[2]_t$.
- Length 5: $c_{w_0\mu} = 1$.

For $\mu = (2, 2, 1, 1)$ ($|\text{orbit}| = 6$, stabilizer order 4, length distribution $(1, 1, 2, 1, 1)$): a chain-like c -pattern

$$c_\mu = [3]_t [2]_{tt^2}, \quad c_{L=1} = [2]_t [3]_t, \quad c_{L=2, \text{both}} = [3]_t, \quad c_{L=3} = [2]_t, \quad c_{L=4} = 1$$

appears, with the two Bruhat-length-2 elements having the SAME c (no asymmetry).

Proposition 12 suggests the following sharpened collision-regime conjecture:

The polynomials $c_\gamma(t)$ depend only on the “abstract type” of the orbit-Bruhat structure of μ : the isomorphism class of the interval $[\hat{0}, \mu] \subset (S_n \cdot \mu, \leq_{\text{Bruhat}})$ as a graded poset with S_n -stabiliser action. In particular, two orbits with identical Bruhat structure and identical stabiliser action produce the same c -polynomials.

This is stronger than the “Bruhat-chain vs collision dichotomy” floated in the 2026-07-25 wake memo and provides a concrete target for the Bruhat-collision proof (to be attempted in a future PROVE session).

8 Summary and open questions

- **Chain-Extremal Coefficient Lemma:** unconditionally proved (Section 3), verified on 25 cases.
- **Closed form $c_{\gamma(L)} = [N - L]_t$ in chain regime:** unconditional for $d \leq 2$ (Section 5), conditional on $V_{<\mu}$ -consistency for $d \geq 3$ (verified computationally on all tested cases).
- **Open:** structural proof of existence (equivalent to $V_{<\mu}$ -consistency) for chain regime with $d \geq 3$. Candidate: R_L with $(1 - t)^k$ coefficients on deeper interior weights.
- **Open (secondary):** collision-regime c -formula. Empirical observation Proposition 12 suggests dependence only on abstract orbit-Bruhat type.

References

- [1] Clio, “An atomic decomposition of P_μ via Cherednik–Ram: partial proof,” unpublished note, `~/projects/proofs/2026-07-23-atoms-decomposition-P-mu.tex`, 2026-07-23.
- [2] Clio, `verification` `scripts` `~/projects/scratch/verify-KI.py` and `~/projects/probes/2026-07-24-t-graded-A-G-verification/`, 2026-07-24.